The empirical recurrence for OEIS A233688, recovered from its tail

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Abstract

OEIS A233688 records "Empirical recurrence of order 85" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 4096$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the whole sequence, so the recurrence holds from $n = 86$ — every index at which it can be written down. Its last coefficient is $c_{85} = 74641288378048 \neq 0$, so no further tail satisfies anything shorter; any order-85 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A233688 is "Number of (n+1)X(5+1) 0..3 arrays with every 2X2 subblock having the sum of the squares of the edge differences equal to 10, and no two adjacent values equal.". It has offset 1 and begins

$$3940, 64756, 1095164, 19588516, 350631128, 6324816860, 113995582540, 2057970474012, \ldots$$

The entry states:

> Empirical recurrence of order 85 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 08:08:53, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^\top M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 4096$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 4096$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. ☐

3 Why the question has to be asked of a tail

Here the stated order is already the minimal order of the sequence itself, so no tail has to be taken.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 8192$ exact terms from $n = 1$ on, it returns order 85 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{85} c_i\, a(n-i),$$

c_1	23	c_{23}	−1955076604289871465	c_{45}	−89737259603807531106645545	c_{67}
c_2	270	c_{24}	55201480991501947937	c_{46}	−88132967445764471302078882	c_{68}
c_3	−8557	c_{25}	205379540811018364	c_{47}	154283463386397098813283057	c_{69}
c_4	−16772	c_{26}	−481084634392696394266	c_{48}	116718337151701476835379901	c_{70}
c_5	1324292	c_{27}	137887687043742420154	c_{49}	−224836138924187266788154922	c_{71}
c_6	−1584246	c_{28}	3484273889110150156625	c_{50}	−128552725089609848241432235	c_{72}
c_7	−117359680	c_{29}	−1810870631618006938706	c_{51}	277530563341439580928330816	c_{73}
c_8	331729096	c_{30}	−21104740316768579980913	c_{52}	116099994763898762737358737	c_{74}
c_9	6793028778	c_{31}	15014516071510125500502	c_{53}	−289668925618779031690104809	c_{75}
c_{10}	−26673918099	c_{32}	107443706856155141316197	c_{54}	−83833532953314503864370802	c_{76}
c_{11}	−275465206614	c_{33}	−93804631253526421226860	c_{55}	254942098361059799432197479	c_{77}
c_{12}	1333808838975	c_{34}	−461481026212277279084119	c_{56}	45881976498345802565469297	c_{78}
c_{13}	8156590584269	c_{35}	467544890957420727039818	c_{57}	−188479737577393154134322940	c_{79}
c_{14}	−46950020203582	c_{36}	1676755301181375037622610	c_{58}	−16245044075783608587095189	c_{80}
c_{15}	−180689202791590	c_{37}	−1908926005393818116456960	c_{59}	116464350938940018402858947	c_{81}
c_{16}	1233655022397432	c_{38}	−5162448287977970089762410	c_{60}	591383286625944834403274	c_{82}
c_{17}	3026605021374057	c_{39}	6478115059144726535473760	c_{61}	−59766580431006630965054584	c_{83}
c_{18}	−25061741760183402	c_{40}	13477023632850343232150487	c_{62}	3933124572561664696750900	c_{84}
c_{19}	−38110221267270869	c_{41}	−18433724122546444880544782	c_{63}	25268638567317381408941302	c_{85}
c_{20}	403080014784186152	c_{42}	−29818890581727677934387758	c_{64}	−3248946130908194166348170	
c_{21}	347014810797862732	c_{43}	44222952010654159397898866	c_{65}	−8713306370898381654513448	
c_{22}	−5221716877710659417	c_{44}	55822698119552610239020724	c_{66}	1613746596823706777883611	

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 86$, and it is the recurrence OEIS A233688 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 1$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{85} - c_1 x^{85-1} - \cdots - c_{85}$. Its constant term is $-c_{85} = -74641288378048$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 85 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 85, so $\deg q = 85$, and it is monic. It holds from some index t on. From $\max(t, 1)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 85, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 14 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 85, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 74641288378048.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-85 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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